

# DECISION TREE

## AIM

To implement a simple Decision Tree algorithm using Python to predict whether a student will Pass or Fail based on study hours and attendance.

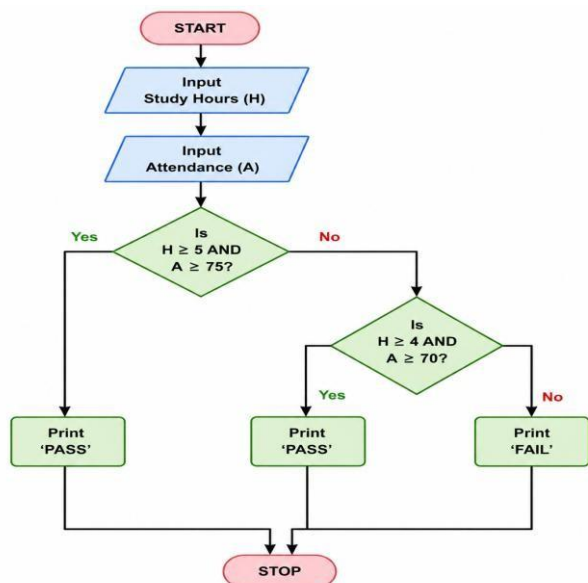
## OBJECTIVE

- To understand the concept of a Decision Tree.
- To use conditions for making decisions.
- To accept study hours and attendance as manual input.
- To predict the student's result.

## ALGORITHM

1. Start the program.
2. Read the student's study hours.
3. Read the student's attendance percentage.
4. Check whether study hours are at least 5 and attendance is at least 75%.
5. If true, display Pass.
6. Otherwise, check whether study hours are at least 4 and attendance is at least 70%.
7. If true, display Pass.
8. Otherwise, display Fail.
9. Stop the program.

## FLOWCHART:

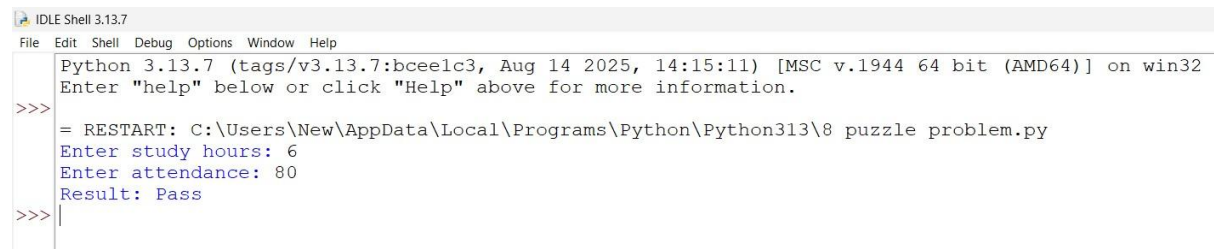


### PYTHON CODE:

```
h = int(input("Enter study hours: "))
a = int(input("Enter attendance: "))

if h >= 5 and a >= 75:
    print("Result: Pass")
elif h >= 4 and a >= 70:
    print("Result: Pass")
else:
    print("Result: Fail")
```

### OUTPUT:



```
IDLE Shell 3.13.7
File Edit Shell Debug Options Window Help
Python 3.13.7 (tags/v3.13.7:bceelc3, Aug 14 2025, 14:15:11) [MSC v.1944 64 bit (AMD64)] on win32
Enter "help" below or click "Help" above for more information.
>>> = RESTART: C:\Users\New\AppData\Local\Programs\Python\Python313\8 puzzle problem.py
Enter study hours: 6
Enter attendance: 80
Result: Pass
>>> |
```

### RESULT:

Thus, the Decision Tree-based program was successfully implemented in Python. It accepts study hours and attendance as manual inputs and predicts whether the student will Pass or Fail.